CURRICULUM VITAE

Yu-Chen Chao (趙于楨)



Accounting and finance graduate · empirical research on digital payments and bank income

New Taipei City, Taiwan · +65 85050726 · chaoglay1101@gmail.com · linkedin.com/in/yu-chen-chao · github.com/chaoglay1101-del

Profile

I am seeking graduate roles in financial data analysis, banking analytics, and FinTech strategy, and master's study in finance or business analytics.

Accounting and finance graduate with a completed empirical research project on digital payment pressure and bank net fee margins in an ASEAN-3 bank panel (39 banks, 344 bank-year observations, 2014–2022). Comfortable across the whole research workflow: building a clean panel dataset, documenting variable definitions, estimating fixed-effects models with clustered standard errors, testing robustness, and communicating what the evidence does and does not support. Additional experience in Tableau data storytelling and cross-border tax work.

Education

DEGREE

BSc (Hons) Accounting and Finance — University of Birmingham

September 2024 – June 2026 · Degree to be conferred 12 November 2026 · Class II (Division I) · Overall weighted mark 61/100 · GPA 3.25

University of Birmingham · Singapore · Campus: Totally Taught Abroad · Language of instruction: English.

DIPLOMA

SIM Diploma in International Business

April 2023 – June 2024

Research experience

2025–2026 · FINAL-YEAR PROJECT

Digital payment pressure and bank net fee margins — ASEAN-3 bank panel

Accounting and Finance Research Project · University of Birmingham

Research report component: 80/100 (65% of module; module total 77/100 on transcript) · above the component's upper quartile (73), i.e. top 25%; cohort mean 67, maximum 84

- Designed the study around an explicit research question and defined the estimation sample: 39 listed commercial banks, 344 bank-year observations, 2014–2022.
- Built a Python pipeline to prepare the bank panel, discover and extract the IMF Financial Access Survey indicator `IMF_FAS_FCMIBT`, and merge the country-year measure into the bank-year panel.
- Constructed `Net_Fee_Margin` and financial controls, winsorised continuous variables at the 1st and 99th percentiles, and formed a complete-case estimation sample.
- Estimated bank and year fixed-effects models with bank-clustered standard errors using `linearmodels`, then tested log, bank-size interaction, nonlinear and country-subsample specifications; checked the result with Driscoll–Kraay errors, a Mundlak decomposition, a one-year lag and a net-interest-margin placebo.
- Reported results as conditional associations, stating the data-level, coverage, and omitted-variable limitations explicitly.
- Published a public research report, variable definitions, and a methodology appendix, plus a runnable synthetic fixed-effects example with an automated test.

Python · pandas · NumPy · linearmodels · statsmodels · matplotlib · pytest · Git and GitHub

BUSINESS ANALYTICS COURSEWORK

Mobile money adoption in the Global South — Tableau story

Business Analytics module · University of Birmingham

Tableau assignment 70/100 (module mark 70/100 on transcript) · equal to the cohort's upper quartile (70); cohort mean 63.7, maximum 76

- Cleaned and documented the dataset in a data-cleaning log, then designed an interactive Tableau story covering adoption trends, regional comparison, and inclusion drivers.
- Presented the findings as a management narrative and published the story on Tableau Public.
- Mapped 437 coded World Bank Findex fields across 174 economies and 12 population groups into 24 business-readable measures, then excluded the 2011 and 2022 waves for coverage, leaving 171 economies and four survey waves (2014–2024).
- Findings: mobile money adoption leads in Sub-Saharan Africa; digital payment use runs ahead of debit-card ownership; savings lag far behind payment usage; and mobile money drives women's financial participation more than traditional banking does.
- Submitted December 2025. Its finding that digital payment use runs ahead of debit-card ownership raised the question my final-year project then tested: what happens to banks' fee income as payments move away from cards.

Tableau | Tableau Story and filters | Tableau calculated fields | Tableau Public | Excel | Python data preparation | World Bank Global Findex

Technical skills

Programming and libraries

- Python 3.14 — pandas, NumPy, linearmodels, statsmodels, matplotlib, openpyxl, requests, BeautifulSoup4, python-docx.
- Reproducible environments with `uv` and dependency lock files; automated tests with `pytest`.
- Version control and collaboration with Git and GitHub, including a continuous-integration workflow.

Econometrics and statistics

- Panel data models with bank and year fixed effects; bank-clustered standard errors.
- Variable transformation and cleaning: log measures, rescaled variables, centred terms, interaction terms, and 1st/99th percentile winsorisation.
- Robustness analysis, subsample estimation, and between-within decomposition.

Data sources and tools

- Bank financial data retrieved through WRDS-derived sources; IMF Financial Access Survey indicators.
- Tableau: dashboards and data stories for non-technical audiences.
- Excel and openpyxl for data preparation, validation, and reconciliation.

Finance and accounting knowledge

- Financial statement analysis, bank profitability and margin analysis, non-interest income.
- Digital payments, FinTech, and financial inclusion.
- Cross-border tax and compliance documentation.

Professional experience

OCTOBER 2026 – PRESENT

Co-founder and technical lead

TechStarterWeb Consulting · web studio for clients in Taiwan, Hong Kong and Singapore

- Built the studio's website and order back office: a public site in Traditional Chinese, Simplified Chinese and English whose enquiry form feeds a signed-in back office for statuses, notes and an activity log.
- Stack: Astro on Cloudflare Workers, a Cloudflare D1 database, Cloudflare Access sign-in with one-time email codes, and OpenCC for the Simplified Chinese pages.
- Ran the launch (domain, redirects, mailbox, Search Console) and wrote the delivery SOPs that split each client project into required and adjustable items.
- Designed how the AI agents work together: a lead agent splits work into task cards, worker agents and a low-cost model carry them out, and scripts and CI check file locks, version numbers and spec updates. From 24 to 27 October 2026, 135 pull requests were opened and 130 merged.
- Built a prototype pipeline: one click in the back office creates a client's repository from a template and publishes a no-index preview; a test order went live in about two and a half minutes.
- Open-sourced the workflow, with tests, as layered-ai-dev-workflow on GitHub.

JULY – SEPTEMBER 2025

Intern, tax advisory team

Evershine CPAs Firm (永輝啟佳聯合會計師事務所)

- Assisted on Philippines tax incentive applications, preparing and checking supporting documentation.
- Mapped standard operating procedures (SOP) and turned them into flowcharts for the application process.
- Deliverable: SOP flowcharts covering the tax incentive application workflow.

Selected projects and links

digital-payment-pressure-bank-margins
Research repository: data workflow, regressions, documentation, and a synthetic example.

mobile-money-adoption-tableau
Tableau workbook and supporting material for the mobile money story.

Tableau Public story
Published interactive visualisation of mobile money adoption.

Research summary on this site
Question, design, results, and limitations in one page.

techstarterweb.com
Studio website and order back office; I co-founded the studio and lead its technical side.

Additional information

Languages

- Chinese (Mandarin) — native.
- English — language of instruction for the degree.

Notes

- Research data restrictions mean the thesis bank panel cannot be published; the public repository therefore provides the documented workflow and a runnable synthetic example instead.
- This CV is generated from the same source as the rest of this site, so it stays consistent with the published research pages.